NAMANGA ELECTRONICS SALES

Complete MS Excel Practical Solution — Step by Step

Based on the uploaded NAMANGA SECONDARY SCHOOL Form Four ICS Spreadsheet Practical Test.

1. Original Data

EmpID	EmployeeName	Department	BasicSalary	UnitsSold	SalesTarget
E001	John Mwakalinga	Electronics	450,000	45	40
E002	Grace Kileo	Household	380,000	28	30
E003	Peter Massawe	Mobile Phones	500,000	62	50
E004	Amina Rashidi	Furniture	420,000	22	25
E006	Neema Mollel	Household	400,000	35	30
E008	Fatuma Juma	Furniture	410,000	30	25

Commission rates: Electronics = 5,000; Household = 3,000; Mobile Phones = 8,000; Furniture = 4,000.

Tax rates: Up to 300,000 = 0%; 300,001–500,000 = 10%; 500,001–700,000 = 20%; Above 700,000 = 30%.

PART (a) — Create and format the worksheet

• Open Microsoft Excel and select Blank Workbook.

• Enter the six headings: EmpID, EmployeeName, Department, BasicSalary (Tsh), UnitsSold and SalesTarget.

• Enter the six employee records exactly as shown in the question.

• Select the heading row, then apply Bold, Wrap Text and Centre Align.

• Go to Page Layout → Orientation → Landscape.

• Save the workbook as NAMANGA ELECTRONICS SALES.

PART (b) — Insert the calculated columns

• Insert seven columns to the right of SalesTarget. Although the question says six new columns, it lists seven: Performance, CommissionRate, Commission, Bonus, GrossPay, Tax and NetPay.

• Your columns should therefore run from A to M: EmpID, EmployeeName, Department, BasicSalary, UnitsSold, SalesTarget, Performance, CommissionRate, Commission, Bonus, GrossPay, Tax, NetPay.

• Format the new headings with Bold, Wrap Text and Centre Align.

PART (c)(i) — CommissionRate

• Click H2 and enter the formula below:

```
=IF(C2="Electronics",5000,IF(C2="Household",3000,IF(C2="Mobile
Phones",8000,IF(C2="Furniture",4000,0))))
```

• Press Enter, then drag/copy H2 down to H7.

PART (c)(ii) — Performance

• Click G2 and enter:

```
=IF(E2>=F2*1.2,"Excellent",IF(E2>=F2,"Good","Poor"))
```

• Press Enter and copy the formula down to G7.

• Expected: John = Good; Grace = Poor; Peter = Excellent; Amina = Poor; Neema = Good; Fatuma = Good.

PART (c)(iii) — Bonus

• Click J2 and enter:

```
=IF(G2="Excellent",D2*10%,IF(G2="Good",D2*5%,0))
```

• Press Enter and copy down to J7.

PART (c)(iv) and (d) — Commission, GrossPay, Tax and NetPay

• Commission in I2:

```
=E2*H2
```

• GrossPay in K2:

```
=D2+I2+J2
```

• Tax in L2:

```
=IF(K2<=300000,0,IF(K2<=500000,K2*10%,IF(K2<=700000,K2*20%,K2*30%)))
```

• NetPay in M2:

```
=K2-L2
```

• Press Enter after each formula and copy each formula down through row 7.

Calculated Results — Check Your Excel

Employee	Performance	Commission	Bonus	GrossPay	Tax	NetPay
John Mwakalinga	Good	225,000	22,500	697,500	139,500	558,000
Grace Kileo	Poor	84,000	0	464,000	46,400	417,600
Peter Massawe	Excellent	496,000	50,000	1,046,000	313,800	732,200
Amina Rashidi	Poor	88,000	0	508,000	101,600	406,400
Neema Mollel	Good	105,000	20,000	525,000	105,000	420,000
Fatuma Juma	Good	120,000	20,500	550,500	110,100	440,400

PART (e) — COUNTIF

```
Excellent: =COUNTIF(G2:G7,"Excellent") → 1
Good: =COUNTIF(G2:G7,"Good") → 3
Poor: =COUNTIF(G2:G7,"Poor") → 2
```

PART (f) — SUMIF Summary

Department	Total NetPay
Electronics	558,000
Household	837,600
Mobile Phones	732,200
Furniture	846,800

```
Use in the total NetPay cell beside the department name: =SUMIF($C$2:$C$7,O2,$M$2:$M$7), then copy down.
```

PART (g) — Conditional Formatting

• Select M2:M7 → Home → Conditional Formatting → Highlight Cells Rules → Greater Than.

• Enter 600000, choose a green fill, and click OK.

• Only NetPay values greater than Tsh 600,000 should be highlighted.

PART (h) — Data Validation

• Select C2:C7 → Data → Data Validation.

• Under Allow choose List.

• Source: Electronics,Household,Mobile Phones,Furniture.

• Under Error Alert, enable the alert, choose Stop, set title to Invalid Department, and message to: Please select a valid department from the drop-down list.

• Click OK and test the drop-down.

PART (i) — Sort and Freeze

• Select the whole table A1:M7.

• Go to Data → Sort.

• Tick My data has headers.

• Sort by NetPay and choose Largest to Smallest.

• Click OK.

• The descending NetPay order is: Peter 732,200; John 558,000; Fatuma 440,400; Neema 420,000; Grace 417,600; Amina 406,400.

• Go to View → Freeze Panes → Freeze Top Row.

PART (j) — 3-D Pie Chart

• Select the SUMIF summary table containing Department and Total NetPay.

• Go to Insert → Pie Chart → 3-D Pie.

• Set the chart title to Net Pay by Department.

• Add Data Labels using Chart Design → Add Chart Element → Data Labels.

PART (k) — Column Chart

• Select EmployeeName, UnitsSold and SalesTarget (B1:B7 and E1:F7; use Ctrl for non-adjacent selections).

• Go to Insert → Column or Bar Chart → 2-D Column → Clustered Column.

• Set the chart title to Sales Performance.

PART (l) — Footer and Print

• Go to Page Layout → Orientation → Landscape.

• Go to Insert → Header & Footer.

• Click the centre footer section and enter your actual Examination Number.

• Go to File → Print and check the print preview.

• Confirm the page is Landscape, the table/charts are positioned correctly, and the examination number appears in the centre footer.

• Click Print.

Formula Quick Reference

Purpose	Formula
Performance	=IF(E2>=F2*1.2,"Excellent",IF(E2>=F2,"Good","Poor"))
CommissionRate	=IF(C2="Electronics",5000,IF(C2="Household",3000,IF(C2="Mobile Phones",8000,IF(C2="Furniture",4000,0))))
Commission	=E2*H2
Bonus	=IF(G2="Excellent",D2*10%,IF(G2="Good",D2*5%,0))
GrossPay	=D2+I2+J2
Tax	=IF(K2<=300000,0,IF(K2<=500000,K2*10%,IF(K2<=700000,K2*20%,K2*30%)))
NetPay	=K2-L2
COUNTIF Excellent	=COUNTIF(G2:G7,"Excellent")
COUNTIF Good	=COUNTIF(G2:G7,"Good")
COUNTIF Poor	=COUNTIF(G2:G7,"Poor")
SUMIF NetPay	=SUMIF($C$2:$C$7,O2,$M$2:$M$7)